

Card 2: Passphrase Lab

The Hackles Kit

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Hackles Lab · Teacher card · Grades 3-12 · [passphrase-lab/index.html](https://quillsup.org/passphrase-lab/index.html)

Learning goal. Every extra random piece multiplies the guessing work, so long and random beats short and clever.

SPECIMEN LABEL	
10 MIN	Dice Vault slider (each die times 6). Compare Secret A (4 real-list words) with Secret B (8 keyboard characters). Read the attacker-model times aloud.
30 MIN	Add the 36-word classroom list and Try it rows from the honest table. Discuss online versus offline guessing, and why reuse and phishing break more accounts than cracking.

Screens. Two builders make practice secrets. The sticky bar and Compare table show the count, bits and times at two models. Two tables reproduce Demo 1 and Lesson 2.

The models, plainly. Classroom model: 1,000 guesses per second. Attacker model: 10 billion per second against a stolen file with a fast hash. Both are teaching assumptions, not measurements. Average means about half the possibilities, longest means all.

Safety. There is nowhere to type: only dice rolls and word pickers. Practice secrets are never for real accounts.

Teach. 3 words is weak, 4 borderline, 5 to 6 strong. Only dice-chosen words count; quotes, pet names and a-to-@ swaps do not. A password manager is the practical answer.

Standards (approximate; verify). CSTA 1B-NI-05 (3-5), 3A/3B-NI cybersecurity (9-12); ISTE 2d; CASEL Responsible Decision-Making.

Accessibility. Labelled native controls, polite live updates, numbers as text, 200 percent zoom. Spanish draft.

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